

Energy Efficient Implementations of Mismatched Filtering for Wearable Ultrasound on the ARM Ethos-U55

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Background, Motivation and Objective

Wearable ultrasound systems have emerged as a promising technology for continuous and non-invasive monitoring of physiological parameters such as blood pressure or heart rate. However, these devices operate under strict power and computational constraints, making efficient on-device signal processing a critical challenge. This work investigates the energy efficiency of mismatched filtering (MMF) implementations on a state-of-the-art microcontroller (MCU; Infineon PSOC-Edge) with an embedded neural processing unit (NPU). For this, MMF is implemented in time- and frequency-domain on the ARM Cortex M55 and the ARM Ethos-U55 NPU of the MCU using different numerical precisions. The implementations are analyzed for energy consumption and normalized mean-squared error (NMSE) using a pre-recorded real-world ultrasound signal with 16-bit fixed-point precision.

Statement of Contribution/Methods

Implementing MMF on the M55 is done via ARM's CMSIS library, which provides convolution- and real-valued Fast Fourier Transformation (RFFT) functions for various bit-depths. For the NPU, TensorFlow-Lite (TFL) models are created. The time-domain models consist of a convolutional layer, while the frequency-domain models require the manual implementation of the (R)FFT and the signal multiplication. On the MCU, runtime is measured with cycle counter events, while the NMSE is measured by comparing the MMF output of the MCU with a ground truth which is calculated on a host computer in NumPy with 64-bit precision. The power consumption is obtained by measuring the voltage drop of the MCU core voltage over a shunt resistor with the Texas Instruments INA219.

Results/Discussion

We show that performing MMF on the NPU requires more power than the M55 implementations. However, the time-domain MMF models on the NPU yield the lowest runtime overall and thus also achieve the lowest energy requirements. In contrast, the frequency-domain models on the NPU do not utilize the accelerator efficiently and thus require increased inference runtime. The NMSE of the frequency-domain methods is significantly higher than for the time-domain models, since the quantized signal is scaled in-between each operation of the (R)FFT on the NPU. Overall, these findings show that utilizing the NPU for signal processing can significantly improve energy efficiency by up to 28% in wearable ultrasound systems.

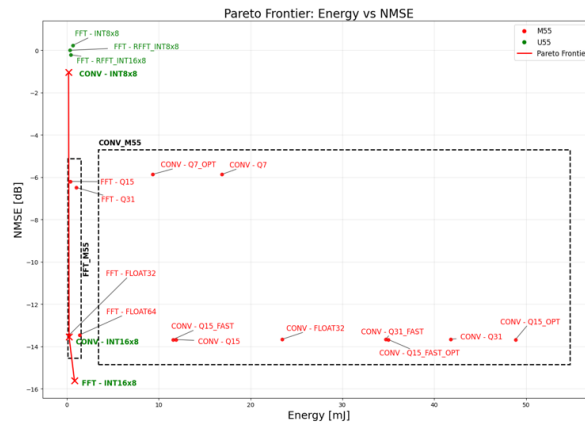


Fig. 1: A Pareto plot showing the trade-offs between energy and NMSE for the investigated MMF methods.